

Congenital Talipes Equino Varus (CTEV)



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Clubfoot



CTEV

CONGENITAL TALIPES EQUINO VARUS

Present at birth

Tali - talus
Pes - foot
“Talus acts as the foot”

Equinus - plantar flexion
(Hindfoot - ankle jt)

Varus - heel inversion
(Subtalar jt)



CTEV vs Clubfoot

CTEV is one of the causes of the clubfoot

Not all clubfoot are caused by CTEV

CTEV

- Males - 2x than females
- Bilateral in 50% of cases
- Genetic inheritance - can affect the siblings
- Look for associated disorders:
Developmental dysplasia of the hip, Spinal problems

Deformities

C - A - V - E

C - CAVUS ; A - ADDUCTUS



Forefoot and Midfoot
(midtarsal joints)

V-VARUS



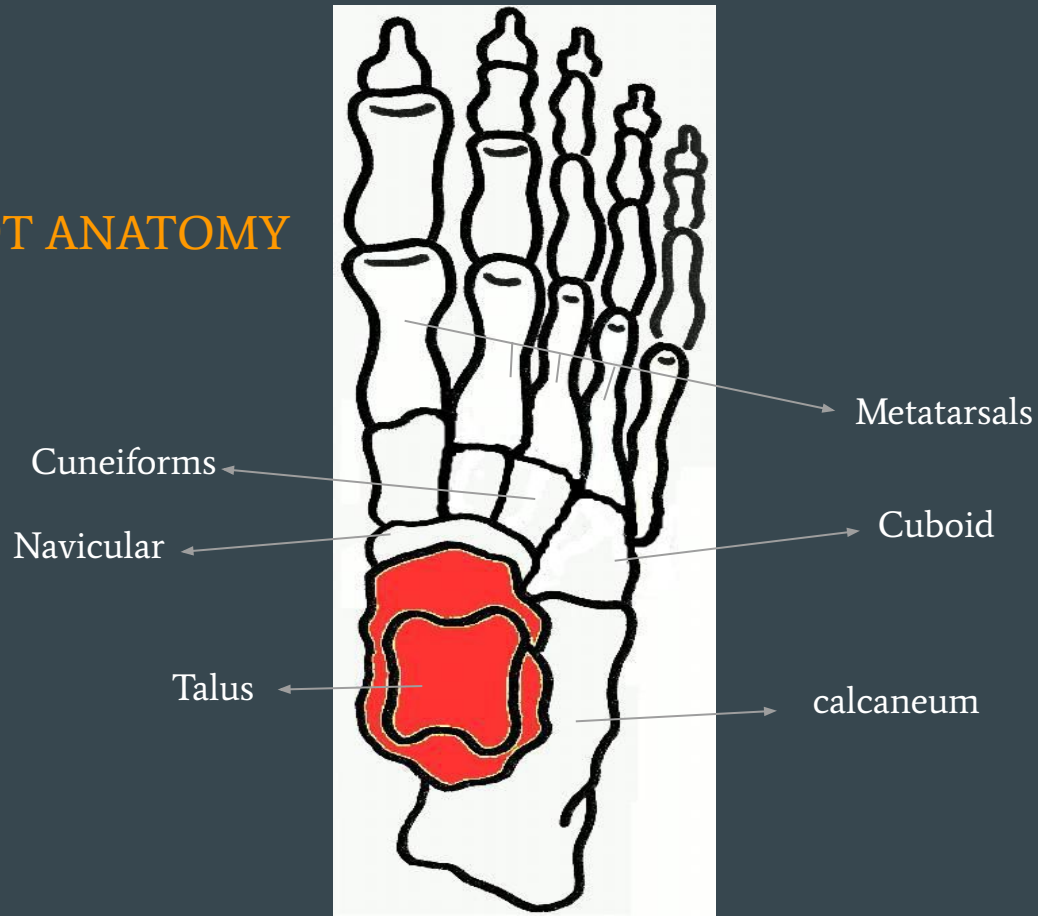
Hindfoot
(subtalar joint)

E-EQUINUS



Hindfoot
(ankle joint)

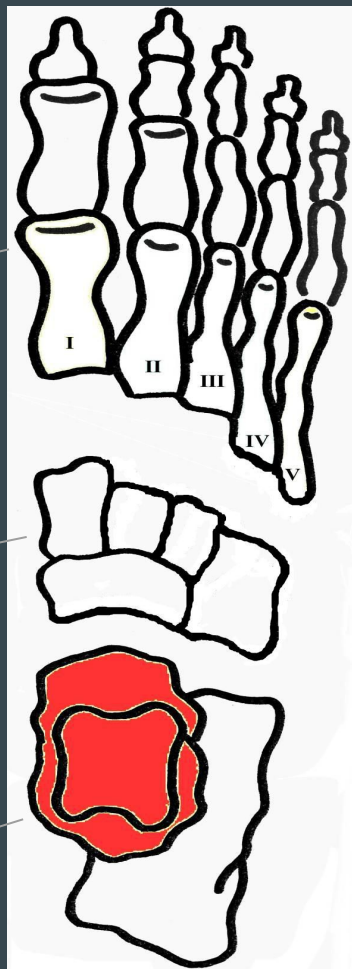
FOOT ANATOMY



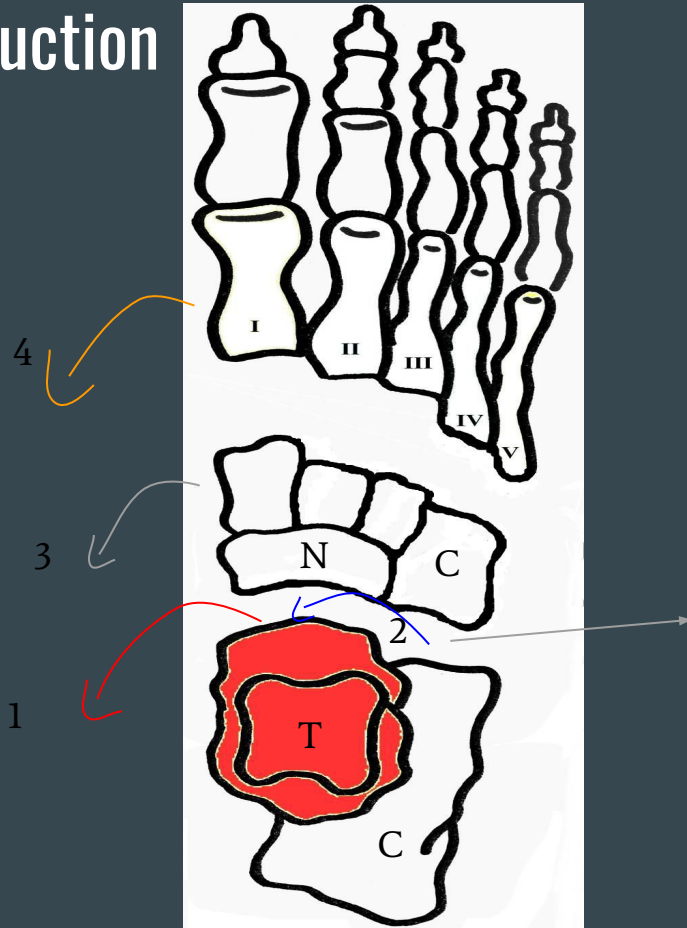
forefoot

midfoot

Hindfoot



Adduction



Midtarsal joints:

1. Talonavicular jt
2. Calcaneocuboid jt

Varus

Posterior view of foot:

medial

Talus

lateral

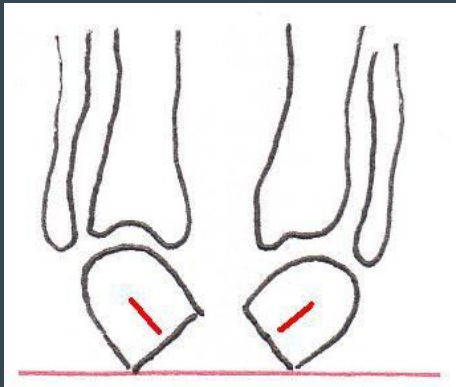
Calcaneum

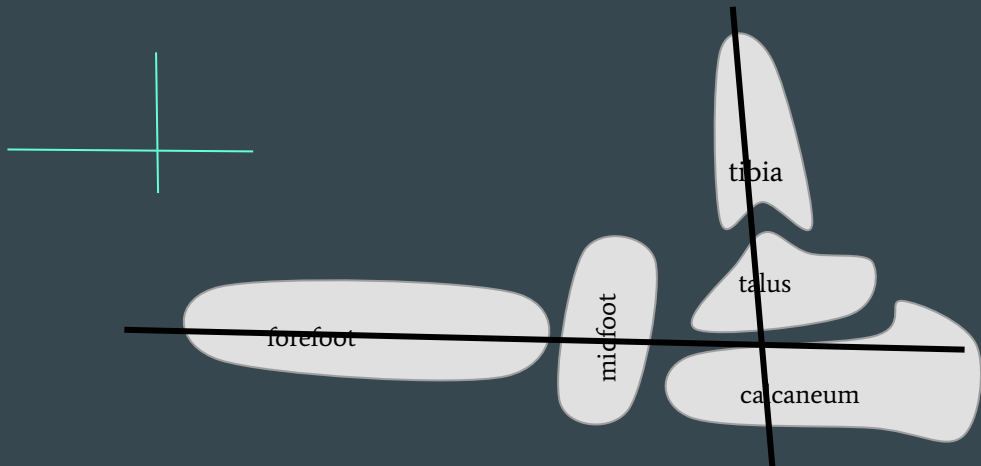
Medial

Lateral

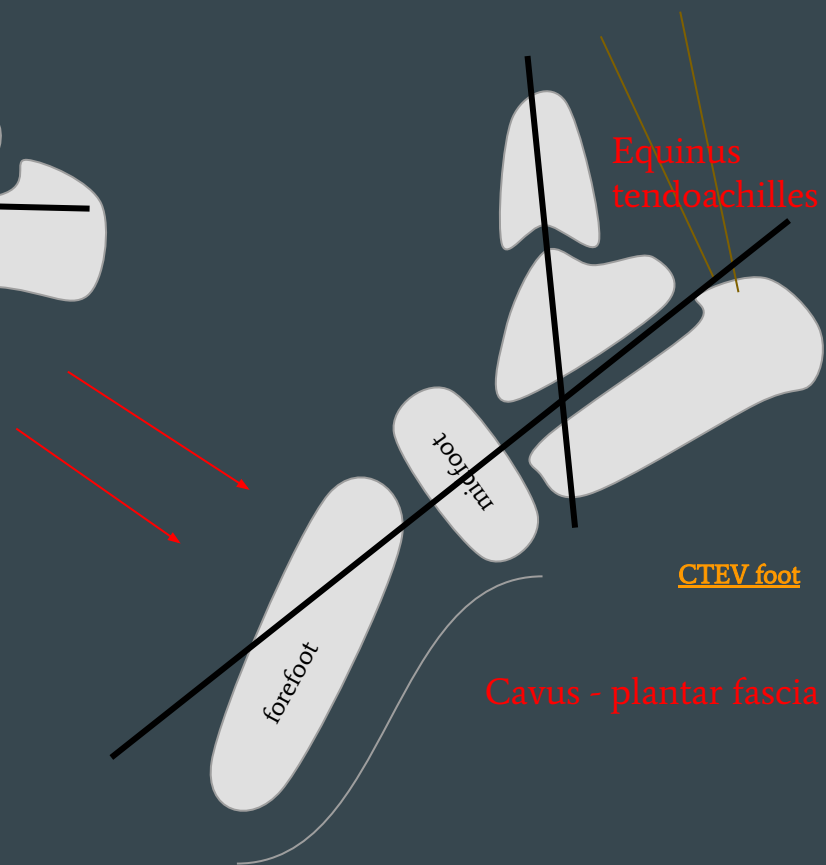
Talus

Calcaneum





Normal Foot

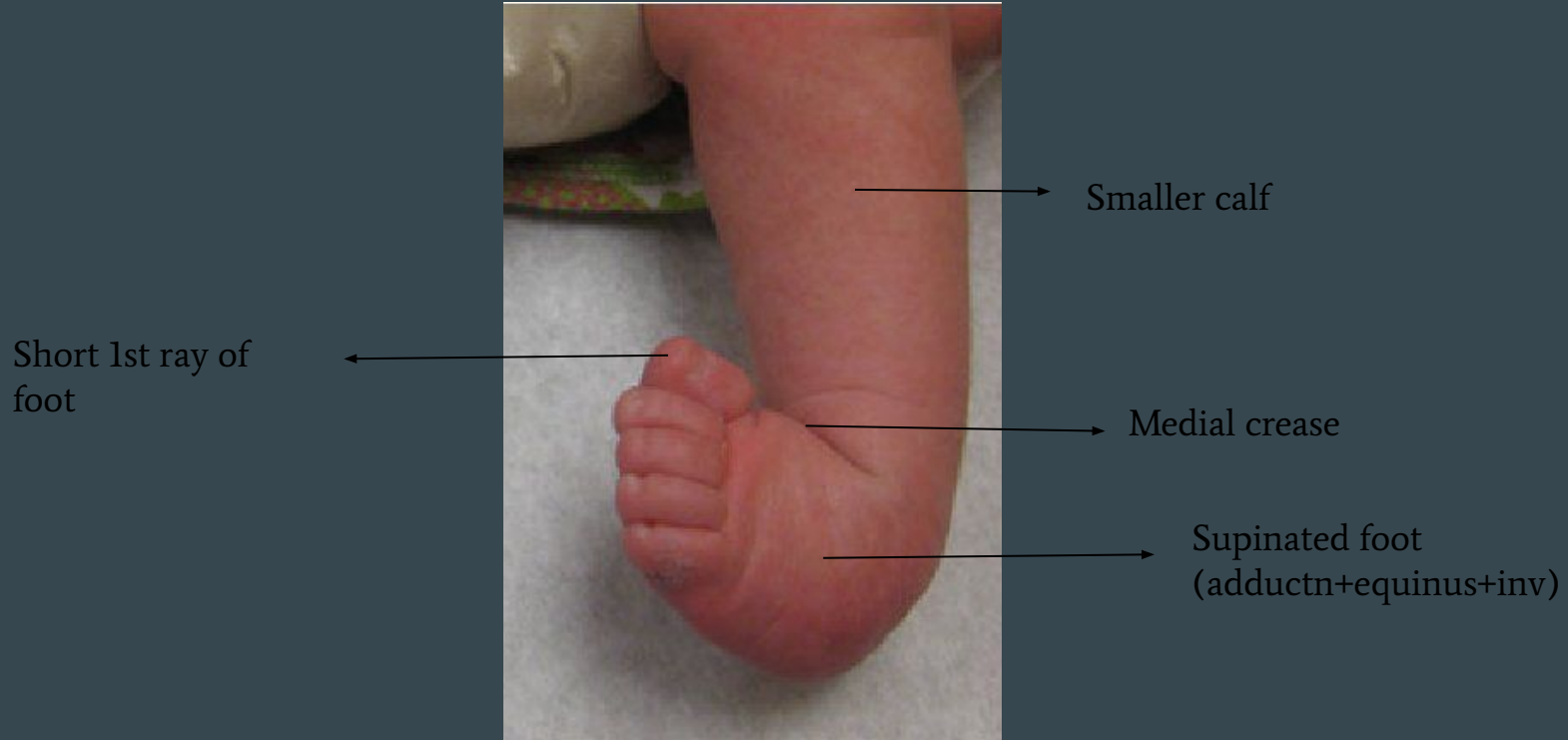


Clinical evaluation

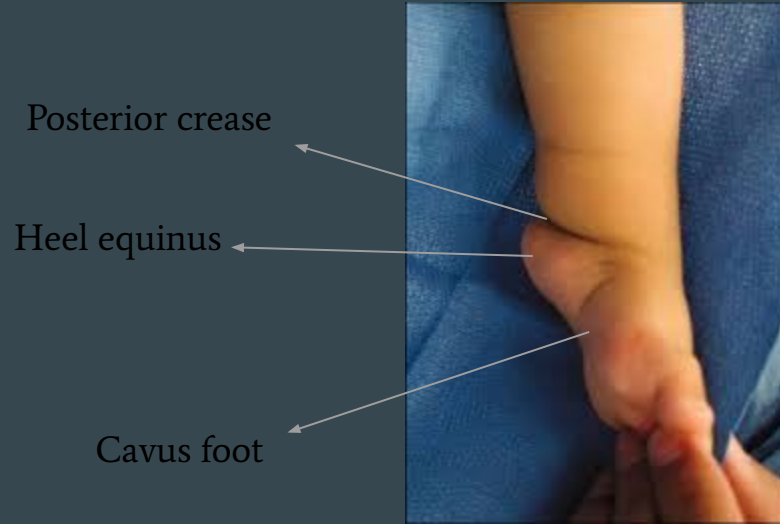
Dorsiflexion test

Dorsiflexion < 20 degrees

Anterior aspect - clinical picture



Lateral aspect



Posterior aspect

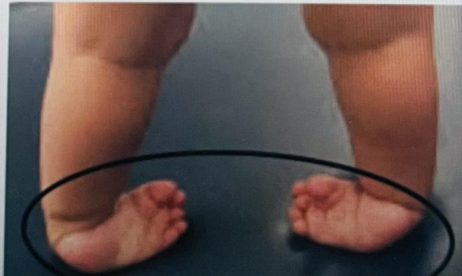
supination

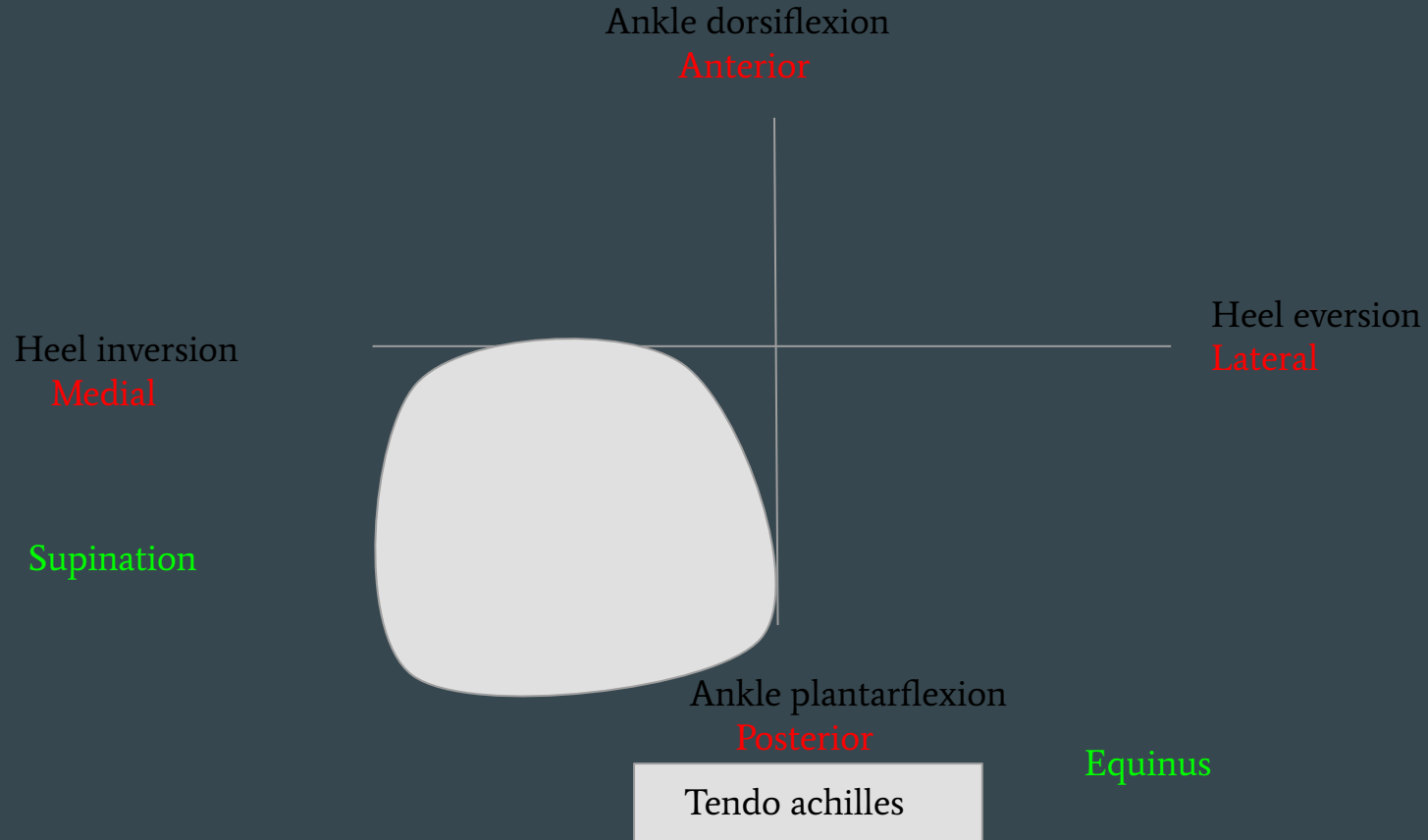


Heel varus

Deformities

HINDFOOT		MIDFOOT
Ankle Joint	Subtalar Joint	Midtarsal Joint
Equinus (Tendo Achilles)	Varus (Tib Post)	Adduction (Tib Post, Abd Hallucis) Cavus (Plantar Fascia)





Management

Goals

1. Plantigrade
2. Mobile
3. Functional
4. Pain free

Treatment methods

Initial treatment - ALWAYS non-surgical

Surgical - scarring and tightness

Principle of treatment - VISCOELASTICITY of soft tissues

Manipulation techniques

“SERIAL MANIPULATION AND CASTING”

PONSETI TECHNIQUE

Sequential correction of deformities

C - A - V - E

1st manipulation and casting

Cavus correction

Dorsiflexing the first ray



Increases the supination
deformity

Groin to toe cast

1st cast - 1 week

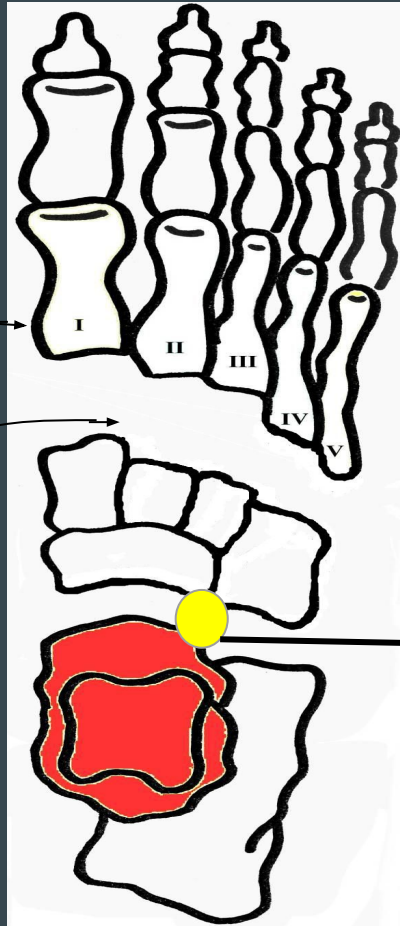


adduction /supination

Supination

Forefoot + midfoot

Normal



2nd cast for 1 week

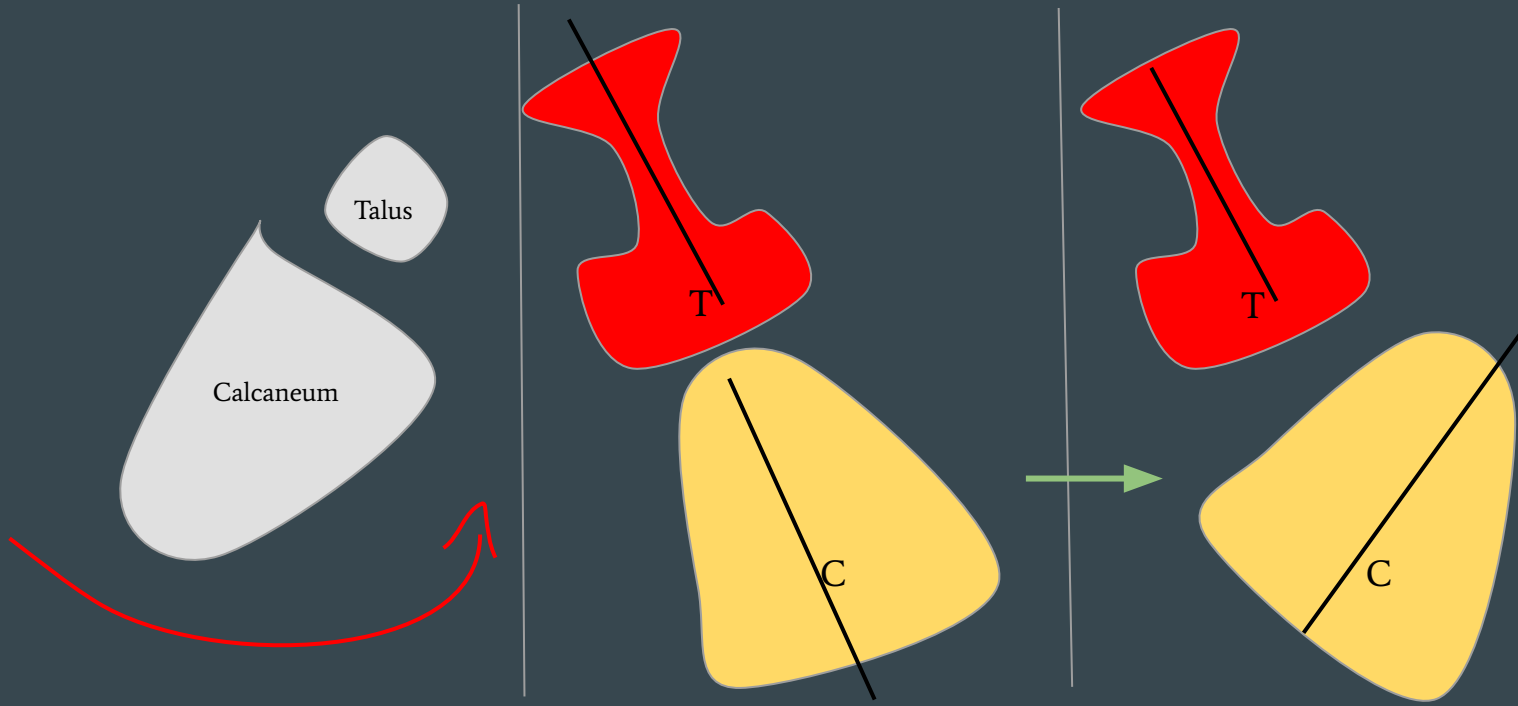
Goal - straight medial border

Fulcrum for
reduction

Varus

VARUS

VALGUS



VIEW FROM BEHIND

FROM ABOVE

Equinus

DORSIFLEXION casting

Percutaneous achilles tenotomy

Maintenance of reduction

Dennis browne splint



Features of a dennis browne shoes

Boot - varus correction

Toe box open - proper growth of the foot

Elevated outer border - supination is prevented

Open heel - to check equinus correction

Straight medial crease - adduction correction



Schedule

DB splint - till walking

DB shoes - after starting to walk

Full time wear - after four months

Night time wear - 3-4 years

Surgical indications

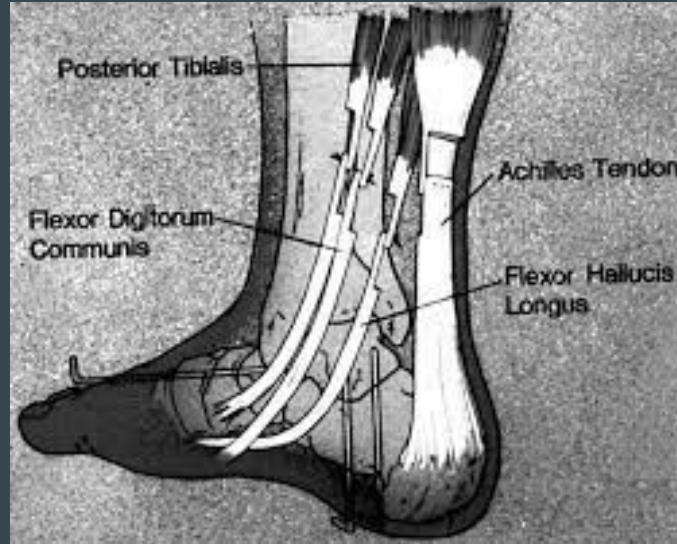
1. Persistent deformity after manipulation
2. Rigid / Resistant
3. Relapse / recurrent

SURGERY

Posteromedial soft tissue release

Preferably done after the walking age (9-10 months)

Posteromedial soft tissue release



Bony procedure

Persistent deformity even after soft tissue procedures.

Congenital clubfoot



Idiopathic clubfoot



Serial manipulation and casting



All deformities corrected



DB splint



equinus persistent



TA lengthening/
Posterior capsule release



all deformities persist



posteromedial soft
tissue release

